

1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Objective

This experiment is divided into four simulations:

1. Generation of Continuous Time Signals
2. Generation of Discrete Time Signals
3. Generation of Triangular and Sawtooth Signals
4. Step and Ramp Functions (Continuous & Discrete)

Program:

```
% Generation of Continuous Time Signals
```

```
clc;
```

```
clear;
```

```
close all;
```

```
% Time axis
```

```
t = -5:0.01:5;
```

```
% Continuous-time sine signal
```

```
x = sin(2*pi*t);
```

```
% Plotting
```

```
figure;
```

```
plot(t, x, 'LineWidth', 2);
```

```
grid on;
```

```
title('Continuous Time Sine Signal');
```

```
xlabel('Time (seconds)');
```

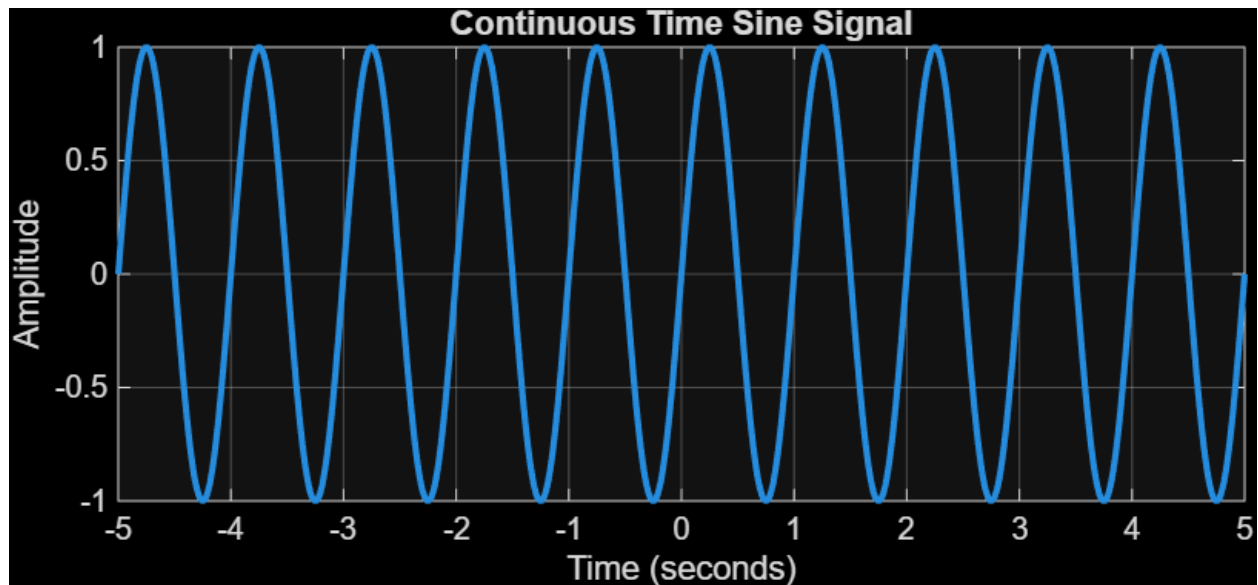
```
ylabel('Amplitude');
```

Expected Output:

A continuous-time sine wave is displayed with:

- Time range: -5 s to 5 s
- Amplitude range: -1 to 1
- Periodic sinusoidal waveform

Output Waveform:



Result:

This test case validates that the signal generator can produce a sinusoidal waveform with the specified frequency, amplitude, and phase shift.

Experiment 1.2 Generation of Discrete Time Signals

Program:

```
% Generation of Discrete-Time Sinusoidal Signal  
clc;  
clear;  
close all;
```

```

% Discrete-time index
n = -10:10;

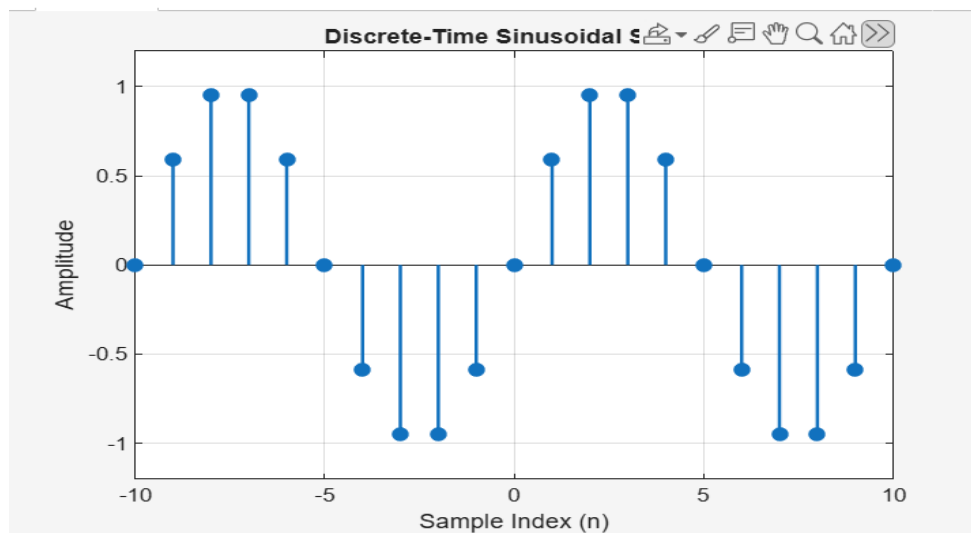
% Discrete-time sinusoidal signal
x = sin(0.2 * pi * n);

% Plot the signal
figure;
stem(n, x, 'filled', 'LineWidth', 1.5);
grid on;
title('Discrete-Time Sinusoidal Signal');
xlabel('Sample Index (n)');
ylabel('Amplitude');
axis([-10 10 -1.2 1.2]);

```

Expected Output:

A discrete-time sinusoidal signal is displayed using stem representation, showing individual samples at discrete time instants.



Result:

This test case validates that the signal generator can accurately produce a discrete-time unit impulse signal at a specified index, which is essential for analyzing the response of discrete time systems.